Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

If $f(\theta) = \sin\theta(\sin\theta + \sin 3\theta)$, then $f(\theta)$

(A) ≥ 0 only when $\theta \geq 0$

(B) ≤ 0 for all real θ

(C) ≥ 0 for all real θ

(D) ≤ 0 only when $\theta \leq 0$

Q.2

If $2\sin^2(x + \pi/4) + \sqrt{3}\cos 2x > 0$, then

(A) $\cos(2x - \pi/6) > -1/2$

(B) $\sin(2x - \pi/6) < -1/2$

(C) $\sin(2x - \pi/6) > -1/2$

(D) $\cos(2x - \pi/6) < -1/2$

Q.3

If A, B, C are the angles of a triangle such that angle A is obtuse then

(A) $\tan A \tan B < 1$

(B) $\tan B \tan C < 1$

(C) $\tan C \tan A < 1$

(D) $\tan A \tan B \tan C < 1$

Q.4

If $\frac{1}{\cos\alpha\cos\beta} + \tan\alpha\tan\beta = \tan\gamma, 0 < \alpha, \beta < \pi$ then $1 - \tan^2\gamma < 0$ for

(A) all values of α and β

(B) no values of α and β

(C) finite number of values of α and β

(D) infinite number of values of α and β

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Q.5

$\frac{\sin 3\alpha}{\cos 2\alpha} < 0$ if α lies in

(A) $(13\pi/48, 14\pi/48)$

(B) $(14\pi/48, 18\pi/48)$

(C) $(18\pi/48, 23\pi/48)$

(D) any of these intervals

Q.6

If $\mathrm{f} : \mathrm{R} \to \mathrm{S}$ is given by $f(x) = \sin x - \sqrt{3}\cos x + 1$ is on to, then the interval of S is

(A) $[0, 1]$

(B) $[-1, 1]$

(C) $[0, 3]$

(D) $[-1, 3]$

Q.7

Statement-1: $1/2 \leq \sin^2\theta + 3\sin\theta\cos\theta + 5\cos^2\theta \leq 11/2$
Statement-2: $-\sqrt{a^2+b^2} \leq a\sin\theta + b\cos\theta \leq \sqrt{a^2+b^2}$.

(A) STATEMENT-1 isTrue, STATEMENT-2 is True; STATEMENT-2 is a correct explanation for STATEMENT-1

(B) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is NOT a correct explanation for STATEMENT-1

(C) STATEMENT-1 is True, STATEMENT-2 is False

(D) STATEMENT-1 is False, STATEMENT-2 is True

Q.8

Let $\theta \in (\pi/4, \pi/2)$, which of the following statements is true?

(A) $(\cos\theta)^{\cos\theta} < (\sin\theta)^{\cos\theta} < (\cos\theta)^{\sin\theta}$

(B) $(\cos\theta)^{\sin\theta} < (\cos\theta)^{\cos\theta} < (\sin\theta)^{\cos\theta}$

(C) $(\sin\theta)^{\cos\theta} < (\cos\theta)^{\cos\theta} < (\cos\theta)^{\sin\theta}$

(D) $(\cos\theta)^{\cos\theta} < (\cos\theta)^{\sin\theta} < (\sin\theta)^{\cos\theta}$

Q.9

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The values of x between 0 and 2π which satisfy the equation $\sin x\sqrt{8\cos^2 x} = 1$ are in A.P. The common difference of the A.P. is

(A) $\pi/8$

(B) $\pi/4$

(C) $3\pi/8$

(D) $5\pi/8$

Q.10

If $D = \begin{vmatrix} 1 & \cos\theta & 1 \\ -\sin\theta & 1 & -\cos\theta \\ -1 & \sin\theta & 1 \end{vmatrix}$ then D lies in the interval

(A) $[0, 4]$

(B) $[2, 4]$

(C) $[2-\sqrt{2}, 2+\sqrt{2}]$

(D) $[-2, 2]$

Q.11

The equation $a\sin x + b\cos x = c$, where $|c| > \sqrt{a^2+b^2}$ has

(A) a unique solution

(B) infinite number of solutions

(C) no solution

(D) none of these

Q.12

If $0 < x < \frac{\pi}{2}$ then

(A) $\cos(\sin x) < \sin(\cos x)$

(B) $\cos(\sin x) < \cos x$

(C) $\cos(\sin x) > \sin(\cos x)$

(D) None

Q.13

If $y = \cos^2 x + \sec^2 x$, then

(A) $y \leq 2$

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(B) $y \leq 1$

(C) $y \geq 2$

(D) $y \geq 1$

Q.14

The set of all x in $(-\pi, \pi)$ satisfying $|4 \sin \mathrm{x}-1|<\sqrt{5}$ is given by

(A) $\left(-\frac{\pi}{10}, \frac{3 \pi}{10}\right)$

(B) $\left(-\frac{\pi}{10}, \pi\right)$

(C) $(-\pi, \pi)$

(D) $\left(-\pi, \frac{3 \pi}{10}\right)$

Q.15

For what values of x is the equation $2 \sin \theta$
$= x+\frac{1}{x}$ valid?

(A) $x= \pm 1$

(B) All real values of x

(C) $-1<\mathrm{x}<1$

(D) $x>1$ and $x<-1$

Answers & Solutions

Q.1 Answer:

≥ 0 for all real θ

Solution:

$f(\theta)=\sin \theta\left(\sin \theta+3 \sin \theta-4 \sin ^3 \theta\right)$
$=4 \sin ^2 \theta\left(1-\sin ^2 \theta\right)=\sin ^2 2 \theta \geq 0$ **for all real** θ

Q.2 Answer:

$\cos (2 x-\pi / 6)>-1 / 2$

Solution:

$2 \sin ^2(x+\pi / 4)+\sqrt{3} \cos 2 x>0$
$\Rightarrow 1-\cos (2 x+\pi / 2)+\sqrt{3} \cos 2 x>0$
$\Rightarrow \frac{1}{2} \sin 2 x+\frac{\sqrt{3}}{2} \cos 2 x>-\frac{1}{2}$
$\Rightarrow \cos (2 x-\pi / 6)>-\frac{1}{2}$

Q.3 Answer:

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$\tan B \tan C < 1$

Solution:

$\because$ **A is obtuse angle.**

$\text{Then } \frac{\pi}{2} < \text{A} < \pi$

$\Rightarrow \frac{\pi}{2} < \pi - (\text{B} + \text{C}) < \pi$

$\Rightarrow -\frac{\pi}{2} > -\pi + (\text{B} + \text{C}) > -\pi$

$\Rightarrow \frac{\pi}{2} > (\text{B} + \text{C}) > 0$

$\because \text{B} + \text{C} < \frac{\pi}{2}$

$\Rightarrow \text{B} < \frac{\pi}{2} - \text{C}$

$\Rightarrow \tan \text{B} < \tan\left(\frac{\pi}{2} - \text{C}\right)$

$\Rightarrow \tan \text{B} < \cot \text{C}$

$\therefore \tan \text{B} < \frac{1}{\tan \text{C}}$

$\text{Hence } \tan \text{B} \tan \text{C} < 1$

Q.4 Answer:

all values of α and β

Solution:

We have $1 - \tan^2 \gamma = \frac{\cos^2 \alpha \cos^2 \beta - (1 + \sin \alpha \sin \beta)^2}{\cos^2 \alpha \cos^2 \beta}$

$= \frac{\left(1 - \sin^2 \alpha\right)\left(1 - \sin^2 \beta\right) - \left(1 + 2 \sin \alpha \sin \beta + \sin^2 \alpha \sin^2 \beta\right)}{\cos^2 \alpha \cos^2 \beta}$

$= \frac{-(\sin \alpha + \sin \beta)^2}{\cos^2 \alpha \cos^2 \beta} < 0 (\sin \alpha + \sin \beta \neq 0, \text{ as } 0 < \alpha, \beta < \pi)$

Q.5 Answer:

$(13\pi/48, 14\pi/48)$

Solution:

$\frac{\sin 3\alpha}{\cos 2\alpha} < 0$ **if** $\sin 3\alpha > 0$ **and** $\cos 2\alpha < 0$

or $\sin 3\alpha < 0$ **and** $\cos 2\alpha > 0$

i.e. if $3\alpha \in (0, \pi)$ **and** $2\alpha \in (\pi/2, 3\pi/2)$

or $3\alpha \in (\pi, 2\pi)$ **and** $2\alpha \in (-\pi/2, \pi/2)$

i.e. if $\alpha \in (0, \pi/3)$ **and** $\alpha \in (\pi/4, 3\pi/4)$

or $\alpha \in (\pi/3, 2\pi/3)$ **and** $\alpha \in (-\pi/4, \pi/4)$

i.e. if $\alpha \in (\pi/4, \pi/3)$

since $(13\pi/48, 14\pi/48) \subset (\pi/4, \pi/3)$, **(a) is correct**

Q.6 Answer:

$[-1, 3]$

Solution:

We have $-\sqrt{a^2 + b^2} \leq a \sin x + b \cos x \leq \sqrt{a^2 + b^2}$

So $-\sqrt{1 + 3} \leq \sin x - \sqrt{3} \cos x \leq \sqrt{1 + 3}$

$\Rightarrow \quad -2 \leq \sin x - \sqrt{3} \cos x \leq 2$

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$\Rightarrow \quad -2+1 \leq \sin x - \sqrt{3}\cos x + 1 \leq 2+1$

$\Rightarrow \quad -1 \leq f(x) \leq 3$

Q.7 Answer:

STATEMENT-1 isTrue, STATEMENT-2 is True; STATEMENT-2 is a correct explanation for STATEMENT-1

Solution:

Take $a = r\cos\alpha$ **and** $b = r\sin\alpha$

$\Rightarrow \quad r = \sqrt{a^2+b^2}$ **and** $a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$

Now $-r \leq r\sin(\theta+\alpha) \leq r.$

$\Rightarrow \quad -\sqrt{a^2+b^2} \leq a\sin\theta + b\cos\theta \leq \sqrt{a^2+b^2}$

$\Rightarrow$ **Statement- 2 is true.**

$$\text{Now} \quad \sin^2\theta + 3\sin\theta\cos\theta + 5\cos^2\theta$$
$$= 1 + (3/2)\sin 2\theta + 4\cos^2\theta$$
$$= 1 + (3/2)\sin 2\theta + 2(1+\cos 2\theta)$$
$$= 3 + (3/2)\sin 2\theta + 2\cos 2\theta = A(\text{ say })$$

From statement-2

$3 - \sqrt{(9/4)+4} \leq A \leq 3 + \sqrt{9/4+4}$

$\Rightarrow \quad 1/2 \leq A \leq 11/2$

and the statement- 1 is also true.

Q.8 Answer:

$(\cos\theta)^{\sin\theta} < (\cos\theta)^{\cos\theta} < (\sin\theta)^{\cos\theta}$

Solution:

For $\theta \in (\pi/4, \pi/2), 0 < \cos\theta < 1/\sqrt{2} < \sin$

$\theta < 1$

$\Rightarrow \quad (\cos\theta)^{\cos\theta} < (\sin\theta)^{\cos\theta}$ **and** $(\cos\theta)^{\cos\theta} > (\cos\theta)^{\sin\theta}$

as $(0 < \cos\theta < 1$ **and** $\cos\theta < \sin\theta)$

Showing that (b) is correct.

Q.9 Answer:

$\pi/4$

Solution:

From the given equation we have

$2\sin x\,|\cos x\,| = 1/\sqrt{2} \Rightarrow \sin 2x = 1/\sqrt{2}$ **if** $\cos x > 0$ **and**

$\sin 2x = -1/2$ **if** $\cos x < 0$

$\therefore \quad$ **when** $\cos x > 0, \sin 2x = 1/\sqrt{2} \Rightarrow x = \pi/8, 3\pi/8$

when $\cos x < 0, \sin 2x = -1/\sqrt{2} \Rightarrow x = 5\pi/8, 7\pi/8$**. So the required values of** x **are**

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$\pi/8, 3\pi/8, 5\pi/8, 7\pi/8$

which form an A.P. with common difference $\pi/4$.

Q.10 Answer:

$[2, 4]$

Solution:

$D = 1 + \sin\theta\cos\theta - \cos\theta(-\sin\theta - \cos\theta) + \left(-\sin^2\theta + 1\right)$

$D = 1 + \sin\theta\cos\theta + \sin\theta\cos\theta + \cos^2\theta + 1 - \sin^2\theta$

$D = 2 + 2\sin\theta\cos\theta + \cos^2\theta - \sin^2\theta$

$D = 2 + \sin 2\theta + \cos 2\theta$

$D = [2, 4]$

Correct Answer

Q.11 Answer:

no solution

Solution:

Let $a = r\cos\alpha, b = r\sin\alpha$ so that

$r = \sqrt{a^2 + b^2}$

The given equation can be written as

$r\sin(x + \alpha) = c \Rightarrow \sin(x + \alpha) = \frac{c}{\sqrt{a^2+b^2}}$

$\Rightarrow \quad |\sin(x + \alpha)| = \frac{|c|}{\sqrt{a^2+b^2}} > 1$ **as** $|c| > \sqrt{a^2 + b^2}$

which is not possible for any value of x.

Q.12 Answer:

$\cos(\sin x) > \sin(\cos x)$

Solution:

Let $f(x) = x - \sin x$

$\Rightarrow f'(x) = 1 - \cos x > 0, \forall x \in (0, \pi/2)$

Thus, $f(x)$ is strictly increasing in $(0, \pi/2)$

$\Rightarrow \mathrm{f(x)} > \mathrm{f(0)}$

$\Rightarrow x - \sin x > 0$

$\Rightarrow x > \sin x$

$\Rightarrow \cos x < \cos(\sin x) \ldots (\mathrm{i})$

Also, for all x in $(0, \pi/2), 0 < \cos x < 1$

$\Rightarrow \cos x < 1 \Rightarrow \cos x > \sin(\cos x) \ldots \ (\mathrm{ii})$

From (i) and (ii), we get,

$\sin(\cos x) < \cos x < \cos(\sin x)$

Q.13 Answer:

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$y \geq 2$

Solution:

Given $y = \cos^2 x + \sec^2 x$

$\Rightarrow \quad y = \cos^2 x + \frac{1}{\cos^2 x} \left(\because \cos x = \frac{1}{\sec x}\right)$

$\Rightarrow \quad y = \cos^2 x + \frac{1}{\cos^2 x} + 2 - 2$

$\Rightarrow y = \left(\cos x - \frac{1}{\cos x}\right)^2 + 2$

$\Rightarrow \quad y = (\cos x - \sec x)^2 + 2$

As $(\cos x - \sec x)^2 = 0$ or positive

$\therefore y = 2$ or $y \geq 2$

Q.14 Answer:

$\left(-\frac{\pi}{10}, \frac{3\pi}{10}\right)$

Solution:

We have $|4 \sin x - 1| < \sqrt{5}$

$\Rightarrow -\sqrt{5} < 4 \sin x - 1 < \sqrt{5}$

$\Rightarrow -\left(\frac{\sqrt{5}-1}{4}\right) < \sin x < \frac{\sqrt{5}+1}{4}$

$\Rightarrow \sin\left(\frac{-\pi}{10}\right) < \sin x < \sin\left(\frac{3\pi}{10}\right)$

$\Rightarrow \mathrm{x} \in \left(-\frac{\pi}{10}, \frac{3\pi}{10}\right) \quad [\because \mathrm{x} \in (-\pi, \pi)]$

Q.15 Answer:

$x = \pm 1$

Solution:

Given : $2 \sin\theta = x + \frac{1}{x}$

We know that $-1 \leq \sin\theta < 1, -2 \leq 2 \sin\theta < 2$

So, $-2 \leq x + \frac{1}{x} < 2$

Thus, the given equation is valid only if $x = \pm 1$

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